

$$\begin{cases} x_1 + 2x_2 + 3x_3 + 4x_4 = 10 \\ 7x_1 + 0x_2 - 5x_3 + 1x_4 = 3 \\ 6x_1 + 1x_2 + 0x_3 - 2x_4 = 5 \\ 8x_1 + 3x_2 - 4x_3 + 1x_4 = 8 \end{cases}$$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 7 & 0 & -5 & 1 \\ 6 & 1 & 0 & -2 \\ 8 & 3 & -4 & 1 \end{pmatrix} \begin{pmatrix} 10 \\ 3 \\ 5 \\ 8 \end{pmatrix} \xrightarrow{\begin{matrix} -7I \\ -6I \\ -8I \end{matrix}} \sim$$

$$\sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & -14 & -26 & -27 \\ 0 & -11 & -18 & -26 \\ 0 & -13 & -28 & -31 \end{pmatrix} \begin{pmatrix} 10 \\ -67 \\ -55 \\ -72 \end{pmatrix} \xrightarrow{-14} \sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & \frac{13}{7} & \frac{27}{14} \\ 0 & -11 & -18 & -26 \\ 0 & -13 & -28 & -31 \end{pmatrix} \begin{pmatrix} 10 \\ \frac{67}{14} \\ -55 \\ -72 \end{pmatrix} \xrightarrow{\begin{matrix} +11II \\ +13II \end{matrix}} \sim$$

$$\sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & \frac{13}{7} & \frac{27}{14} \\ 0 & 0 & \frac{13}{7} & -\frac{67}{14} \\ 0 & 0 & -\frac{27}{7} & -\frac{137}{14} \end{pmatrix} \begin{pmatrix} 10 \\ \frac{67}{14} \\ -\frac{33}{14} \\ -\frac{137}{14} \end{pmatrix} \xrightarrow{\cdot \frac{7}{17}} \sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & \frac{13}{7} & \frac{27}{14} \\ 0 & 0 & 1 & -\frac{67}{34} \\ 0 & 0 & -\frac{27}{7} & -\frac{83}{14} \end{pmatrix} \begin{pmatrix} 10 \\ \frac{67}{14} \\ -\frac{33}{34} \\ -\frac{137}{14} \end{pmatrix} \xrightarrow{+ \frac{27}{7} III} \sim$$

$$\sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & \frac{13}{7} & \frac{27}{14} \\ 0 & 0 & 1 & -\frac{67}{34} \\ 0 & 0 & 0 & -\frac{230}{17} \end{pmatrix} \begin{pmatrix} 10 \\ \frac{67}{14} \\ -\frac{33}{34} \\ -\frac{230}{17} \end{pmatrix} \xrightarrow{\cdot (-\frac{17}{230})} \sim$$

$$\sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & \frac{13}{7} & \frac{27}{14} \\ 0 & 0 & 1 & -\frac{67}{34} \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 10 \\ \frac{67}{14} \\ -\frac{33}{34} \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{cases} x_1 + 2x_2 + 3x_3 + 4x_4 = 10 \\ x_2 + \frac{13}{7}x_3 + \frac{27}{14}x_4 = \frac{67}{14} \\ x_3 - \frac{67}{34}x_4 = -\frac{33}{34} \\ x_4 = 1 \end{cases}$$

$$x_4 = 1$$

$$x_3 = -\frac{33}{34} + \frac{67}{34} \cdot x_4 = -\frac{33}{34} + \frac{67}{34} \cdot 1 = \frac{34}{34} = 1$$

$$x_2 = \frac{67}{14} - \frac{27}{14}x_4 - \frac{13}{7}x_3 = \frac{67}{14} - \frac{27}{14} \cdot 1 - \frac{26}{14} \cdot 1 = \frac{14}{14} = 1$$

$$x_1 = 10 - 4x_4 - 3x_3 - 2x_2 = 10 - 4 \cdot 1 - 3 \cdot 1 - 2 \cdot 1 = 1$$

$$\text{Ответ: } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$